

Fundamental Stucture (2)

(기본 구조 (2))

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Today's Topics

1. Poisson Regression

- Loglinear model
- Overdispersion
- Quasi-likelihood
- Offset

2. Estimation of Parameters

3. Inference and Testing for Models

4. Deviance

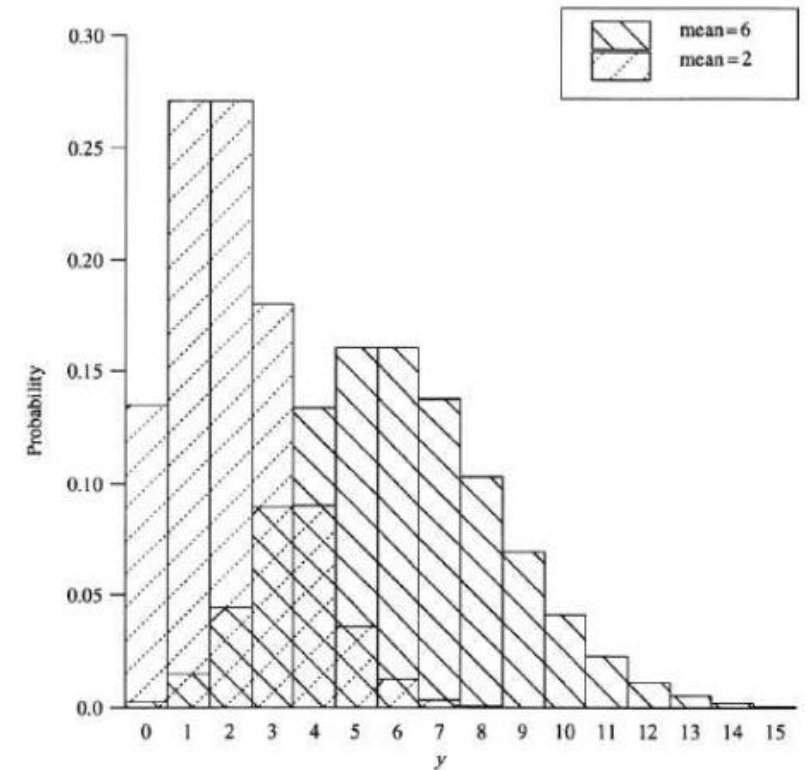
Poisson Regression

When the response variable consists of counts

Examples: number of traffic accidents, number of breakdowns, etc.

Non-negative integers – possible probability models: Poisson, geometric, negative binomial distributions

Assume the Poisson distribution as the random component: The Poisson distribution has the property that as the mean increases, the variance also increases



Poisson Regression (Cont'd)

Poisson Regression

The Poisson mean is non-negative, but the linear predictor can take any real value

The log mean is the natural parameter, and the log link is the canonical link function

Poisson loglinear model

Distribution of Y : Poisson

Link function: log link

$$\log(\mu) = \alpha + \beta x \rightarrow \mu = \exp(\alpha + \beta x) = e^\alpha e^{\beta x}$$

When x increases by 1, the mean increases by a factor of e^β

Poisson Regression (Cont'd)

(예제) Horseshoe Crabs and their satellites

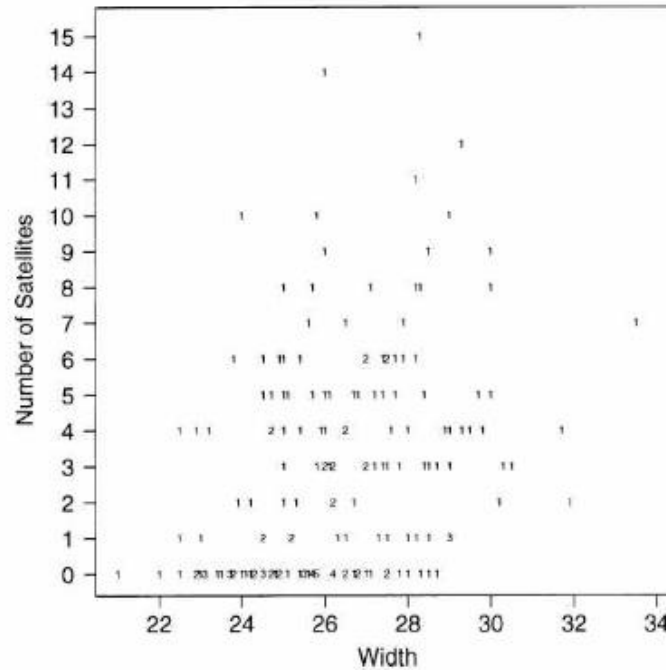


Figure 3.4. Number of satellites by width of female crab.

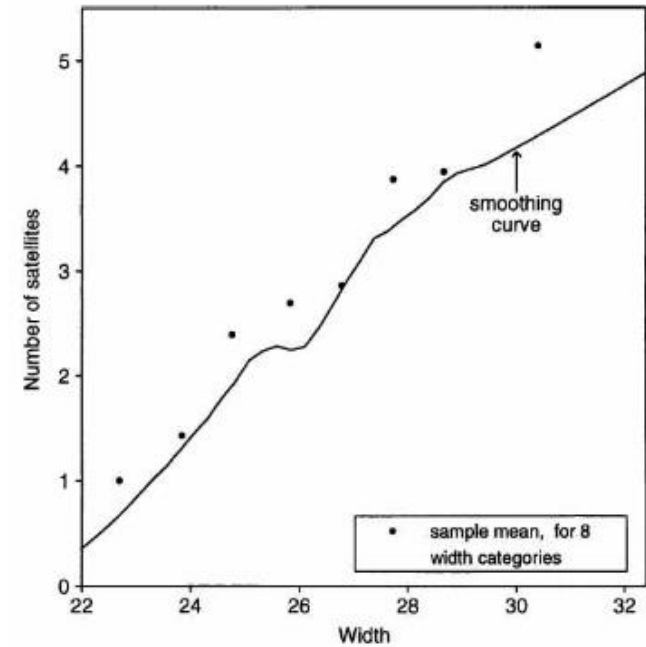


Figure 3.5. Smoothings of horseshoe crab counts.

Poisson Regression (Cont'd)

Model

Width is divided into the following intervals: $(-\infty, 23.25]$, $(23.25, 24.25]$, $(24.25, 25.25]$, $(25.25, 26.25]$, $(26.25, 27.25]$, $(27.25, 28.25]$, $(28.25, 29.25]$, $(29.25, 30.25]$, $(30.25, \infty)$

The sample mean of the scores within each interval of width is used

Fitting Results

$\log \hat{\mu}_i = \hat{\alpha} + \hat{\beta}x_i$, where $E(Y_i|x_i) = \mu_i$, μ_i : mean of satellites, x_i : width

$\hat{\alpha} = -3.305$, $\hat{\beta} = 0.164$, $SE(\hat{\beta}) = 0.020$

$\exp(\hat{\beta}) = 1.18$: When x increases by 1 unit, the expected count increases by a factor of 1.18

Overdispersion

Overdispersion (과분산): greater variability than expected

The Poisson distribution assumes that, for each cell, the mean and variance are equal

If this assumption is violated, overdispersion is said to occur

Crab data example

Width	N of Cases	N of Satellites	Sample Mean	Sample Variance
$(-\infty, 23.25]$	14	14	1.00	2.77
$(23.25, 24.25]$	14	20	1.43	8.88
$(24.25, 25.25]$	28	67	2.39	6.54
$(25.25, 26.25]$	39	105	2.69	11.38
$(26.25, 27.25]$	22	63	2.86	6.88
$(27.25, 28.25]$	24	93	3.87	8.81
$(28.25, 29.25]$	18	71	3.94	16.88
$(29.25, \infty]$	14	72	5.14	8.29

Overdispersion (Cont'd)

General cause of overdispersion: heterogeneity

In the case of the crab data, the distribution may follow different Poisson means depending on gender

If such variables are not accounted for, each cell becomes a mixture of Poisson distributions, and therefore the mean and variance differ

Solution

1. Negative Binomial Regression

(i) Two-stage hierarchical process

$$Z \sim \Gamma(\theta, \theta) \text{ and } Y|Z \sim \text{Poisson}(\lambda Z) \Rightarrow Y \sim \text{NB}$$

(ii) Convergence of negative binomial to Poisson

$$\lim_{r \rightarrow \infty, \frac{r(1-p)}{p} = \text{constant}} \text{NB}(r, p) = \text{Poisson}\left(\frac{r(1-p)}{p}\right)$$

Quasi-likelihood

2. Quasi-Poisson Likelihood

For the Poisson distribution, $\psi = 1$, but assume $\psi \neq 1$ and estimate ψ

Quasi-log likelihood

$$QL = \int_y^\mu \frac{y-t}{a(\psi) V(t)} dt$$

It should be noted that

$$\frac{\partial l}{\partial \mu} = \frac{\partial l}{\partial \theta} \frac{\partial \theta}{\partial \mu} = \frac{y - b'(\theta)}{a(\psi)} \frac{1}{b''(\theta)} = \frac{y - \mu}{a(\psi) V(\mu)}$$

Quasi-likelihood (Cont'd)

Properties: If we let

$$U = \frac{y - \mu}{a(\psi)V(\mu)}$$

1. $E(U) = 0$

2. $var(U) = \frac{1}{a(\psi)V(\mu)} \cdot E\left(-\frac{\partial U}{\partial \mu}\right) = \frac{1}{a(\psi)V(\mu)}$

Offset

Poisson Regression for Rate Data

When events occur relative to some measure of size such as time, space, or another denominator

(Example) Accident rate = number of traffic accidents / number of vehicles

When the index of the response count Y is t , the standardized response is Y/t :

$$E(Y/t) = \mu/t$$

$$\log(\mu/t) = \log(\mu) - \log(t) = \alpha + \beta x$$

$$\mu = t \exp(\alpha + \beta x)$$

The expected value is proportional to t

Offset: $\log(t)$

Offset (Cont'd)

(Example) British Train Accidents over Time between 1975 and 2003

Y : annual number of collisions between trains and road vehicles for t million km of train travel

X : number of years since 1975

<Collisions involving trains in Great Britain>

Year	Train-km	Train Collisions	Train-road Collisions	Year	Train-km	Train Collisions	Train-road Collisions
2003	518	0	3	1988	443	2	4
2002	516	1	3	1987	397	1	6
2001	508	0	4	1986	414	2	13
2000	503	1	3	1985	418	0	5
1999	505	1	2	1984	389	5	3
1998	487	0	4	1983	401	2	7
1997	463	1	1	1982	372	2	3
1996	437	2	2	1981	417	2	2
1995	423	1	2	1980	430	2	2
1994	415	2	4	1979	426	3	3
1993	425	0	4	1978	430	2	4
1992	430	1	4	1977	425	1	8
1991	439	2	6	1976	426	2	12
1990	431	1	2	1975	436	5	2
1989	436	4	4				

Source: British Department of Transport.

Offset (Cont'd)

(Fitted Result) t : million km of train level

$$\log(\hat{\mu}/t) = -4.21 - 0.329x \Leftrightarrow \hat{\mu} = t \exp(-4.21 - 0.329x)$$

Interpretation: For the year 2003, $x = 28$. As this time, $\exp(-4.21 - 0.329x) = 0.0059$. Therefore, there are 0.0059 collisions per million km

Estimation of Parameters

Sufficient statistics for regression coefficients

Equation: Regression coefficients are the root of the following equation:

$$\sum_{i=1}^n \frac{x_i (y_i - \mu_i)}{a(\psi) b''(\theta_i)} \frac{\partial \mu_i}{\partial \eta_i} = 0$$

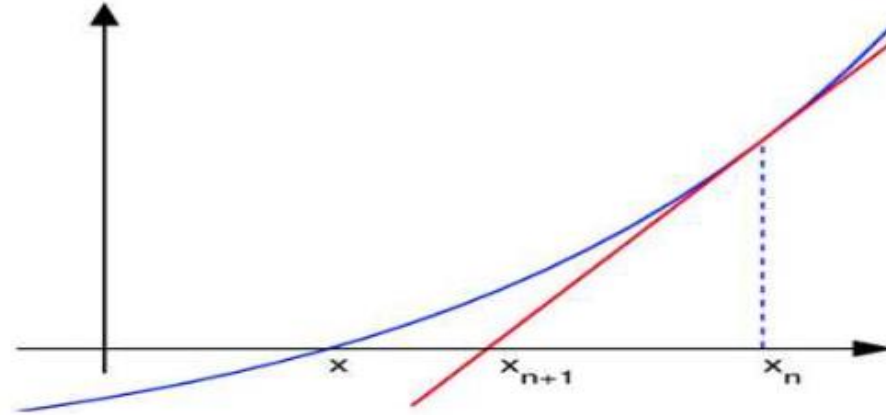
Algorithm for estimating regression coefficients

Newton-Raphson Algorithm

$$U(\beta) \equiv \sum_{i=1}^n \frac{x_i (y_i - \mu_i)}{a(\psi) b''(\theta_i)} \frac{\partial \mu_i}{\partial \eta_i}$$

$$U(\beta) = 0 \Rightarrow \beta^{(m)} = \beta^{(m-1)} - \frac{U(\beta^{(m-1)})}{U'(\beta^{(m-1)})}$$

Estimation of Parameters (Cont'd)



Newton-Raphson Algorithm

$$\hat{\beta}^{(m)} = \hat{\beta}^{(m-1)} - \left(\frac{\partial^2 l}{\partial \beta \partial \beta^t} \right)^{-1} \left[\frac{\partial l}{\partial \beta^t} \right]_{\beta = \hat{\beta}^{(m-1)}}$$

Scoring Algorithm

$$\hat{\beta}^{(m)} = \hat{\beta}^{(m-1)} - \left(I_{\beta = \hat{\beta}^{(m-1)}} \right)^{-1} \left[\frac{\partial l}{\partial \beta^t} \right]_{\beta = \hat{\beta}^{(m-1)}}$$

Estimation of Parameters (Cont'd)

Property:

The scoring algorithm is a modified iterative weighted least square method

When the canonical link function is used, the scoring algorithm and the Newton–Raphson algorithm are identical

If $a(\psi)$ is a constant, ψ can be ignored when estimating regression coefficients

The variance–covariance matrix of the MLE is given by $(X^t W X)^{-1}$, where

$$W = \text{diag}(V(\mu_1), \dots, V(\mu_n))$$

Inference and Testing for the Model

Hypothesis for regression coefficient β :

$$H_0: \beta = 0 \text{ vs } H_1: \beta \neq 0$$

Wald test, Likelihood ratio test, score test

Wald Test

Uses the asymptotic normality of the maximum likelihood estimator

$$Z = \hat{\beta} / ASE$$

Wald statistic: $Z^2 \sim \chi_1^2$

Inference and Testing for the Model (Cont'd)

Likelihood Ratio Test

Ratio of the maximum likelihood under the null hypothesis parameter space to that under the full parameter space

$$-2 \log(L_0/L_1) = -2[\log(L_0) - \log(L_1)] \sim \chi_1^2$$

Score Test

Score function:
$$S(0) = \left[\frac{\partial \log(L(\beta))}{\partial \beta} \right]_{\beta=0} \sim N(0, I_n(\beta=0))$$

Score test statistic:
$$\frac{S(0)^2}{I_n(0)} \sim \chi^2(df=1)$$

Inference and Testing for the Model (Cont'd)

Wald vs LR vs Score test

$$L(\beta) \equiv \log f(Y|\beta): H_0: \beta = 0 \text{ vs } H_1: \beta \neq 0$$

$$\text{Wald Test: } \hat{\beta} / \sqrt{I(\hat{\beta})}$$

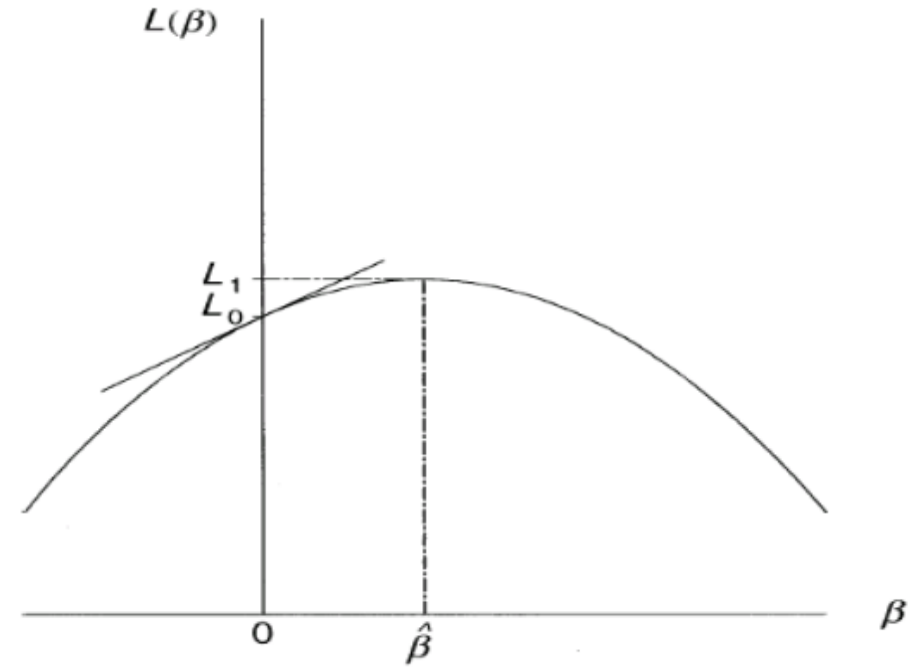
Rejects the null hypothesis when $\hat{\beta}$ is sufficiently far from 0

$$\text{Score test: } l'(0) / \sqrt{I(\beta = 0)}$$

Reject the null hypothesis when the slope of the tangent at $l(0)$ differs significantly from 0

LR test:

Reject the null hypothesis when the difference between $l(0)$ 와 $l(\hat{\beta})$ is large



Deviance

Definition

l_M : maximized log-likelihood value for a model M

l_S : maximized log-likelihood value for the most complex model

Deviance: $-2a(\psi)(l_M - l_S) = -2 \sum_{i=1}^n \left(y_i(\hat{\theta}_{M,i} - \hat{\theta}_{S,i}) - (b(\hat{\theta}_{M,i}) - b(\hat{\theta}_{S,i})) \right)$

Scaled Deviance: $-2(l_M - l_S) = \frac{\text{Deviance}}{a(\psi)}$

Deviance (Cont'd)

(Example) Snoring and Heart Disease

Snoring	Heart Disease		Proportion Yes	Linear Fit	Logit Fit	Probit Fit
	Yes	No				
Never	24	1355	0.017	0.017	0.021	0.020
Occasional	35	603	0.055	0.057	0.044	0.046
Nearly every night	21	192	0.099	0.096	0.093	0.095
Every night	30	224	0.118	0.116	0.132	0.131

Note: Model fits refer to proportion of “yes” responses.

Source: P. G. Norton and E. V. Dunn, *Br. Med. J.*, 291: 630–632, 1985, published by BMJ Publishing Group.

$$l_M = \pi(x) = \alpha + \beta x, \quad x: \text{snoring}, \quad l_S: \pi(x) = \pi_1, \pi_2, \pi_3 \text{ or } \pi_4$$

$$-2(l_M - l_S) = 6.60 \times 10^{-6} \sim \chi^2(df = 4 - 2 = 2), \quad P\text{-value} = 1$$

Interpretation: There is no significant difference in the explanatory power between the two models. That is, using l_M does not cause a problem

Deviance (Cont'd)

Model Comparison using Deviance

For two models M_0 and M_1 , we assume that M_0 is a special case of M_1

LRT: $-2(l_{M_0} - l_{M_1}) \sim \chi^2(df = \text{difference in number of parameters})$

$$-2(l_{M_0} - l_{M_1}) = Dev_0 - Dev_1$$

Residuals Comparing Observations to the Model Fits

Pearson residual: $e_i = \frac{y_i - \hat{\mu}_i}{\sqrt{\text{var}(y_i)}}$

Adjusted residual: $e_i = \frac{y_i - \hat{\mu}_i}{\sqrt{\text{var}(y_i - \hat{\mu}_i)}}$

Deviance residual: $e_i = \text{sign}(y_i - \hat{\mu}_i) \sqrt{-2a(\psi)(l_{M,i} - l_{S,i})}$

R Code

코골이	Score	심장질환		비율	선형	로짓	프로빗
		예	아니오				
전혀 안함	0	24	1355	0.017	0.017	0.021	0.020
가끔씩	2	35	603	0.055	0.057	0.044	0.046
거의 매일	4	21	192	0.099	0.096	0.093	0.095
매일	5	30	224	0.118	0.116	0.132	0.131

```
res1 <-  
data.frame(snore,p_hat_lm=fit_lm$fitted.values,p_hat_logit=fit  
_logit$fitted.values,  
p_hat_probit=fit_probit$fitted.values,p_hat_cll=fit_cll$fitted.val  
ues)
```

```
## Snoring and Heart disease  
snore <- data.frame(snore_score = c(0, 2, 4, 5), yes = c(24, 35, 21, 30),  
no = c(1355, 603, 192, 224))  
snore$trials <- snore$yes + snore$no  
snore$prop <- snore$yes / snore$trials
```

```
# Identity  
fit_lm <- lm(prop ~ snore_score, data=snore)  
summary(fit_lm)
```

```
# Logit  
fit_logit <- glm(cbind(yes, no) ~ snore_score, data=snore,  
family=binomial(link="logit"))  
summary(fit_logit)
```

```
# Probit  
fit_probit <- glm(cbind(yes, no) ~ snore_score, data=snore,  
family=binomial(link="probit"))  
summary(fit_probit)
```

```
# Complementary log-log  
fit_cll <- glm(cbind(yes, no) ~ snore_score, data=snore,  
family=binomial(link="cloglog"))  
summary(fit_cll)
```

R Code (Cont'd)

코골이	Score	심장질환		비율	선형	로짓	프로빗
		예	아니오				
전혀 안함	0	24	1355	0.017	0.017	0.021	0.020
가끔씩	2	35	603	0.055	0.057	0.044	0.046
거의 매일	4	21	192	0.099	0.096	0.093	0.095
매일	5	30	224	0.118	0.116	0.132	0.131

```
# Deviance
fit_lm <- lm(prop ~ snore_score, data=snore)
dev <- deviance(fit_lm); dev
df <- df.residual(fit_lm); df
p_gof <- pchisq(dev, df, lower.tail = FALSE); p_gof

res_pear <- residuals(fit_lm, type="pearson"); res_pear
res_adj <- rstandard(fit_lm, type = "pearson"); res_adj
res_dev <- residuals(fit_lm, type="deviance"); res_dev

fit_logit <- glm(cbind(yes, no) ~ snore_score, data=snore,
family=binomial(link="logit"))
res_pear <- residuals(fit_logit, type="pearson"); res_pear
res_adj <- rstandard(fit_logit, type = "pearson"); res_adj
res_dev <- residuals(fit_logit, type="deviance"); res_dev
```


R Code (Cont'd)

Log Dose	No. Beetles	No. Killed	Fitted Values		
			Cloglog	Probit	Logit
1.691	59	6	5.7	3.4	3.5
1.724	60	13	11.3	10.7	9.8
1.755	62	18	20.9	23.4	22.4
1.784	56	28	30.3	33.8	33.9
1.811	63	52	47.7	49.6	50.0
1.837	59	53	54.2	53.4	53.3
1.861	62	61	61.1	59.7	59.2
1.884	60	60	59.9	59.2	58.8

```
## Beetles and CS2
beetles <- data.frame(
  logdose = c(1.691,1.724,1.755,1.784,1.811,1.837,1.861,1.884),
  n       = c(59,60,62,56,63,59,62,60),
  killed  = c(6,13,18,28,52,53,61,60)
)
beetles$survive <- beetles$n - beetles$killed
beetles$prop   <- beetles$killed / beetles$n

fit_logit <- glm(cbind(killed, survive) ~ logdose, data=beetles,
family=binomial("logit"))
summary(fit_logit)

fit_probit <- glm(cbind(killed, survive) ~ logdose, data=beetles,
family=binomial("probit"))
summary(fit_probit)

fit_cll <- glm(cbind(killed, survive) ~ logdose, data=beetles,
family=binomial("cloglog"))
summary(fit_cll)

res2 <- data.frame(beetles,p_hat_logit=fit_logit$fitted.values,
p_hat_probit=fit_probit$fitted.values,p_hat_cll=fit_cll$fitted.values)
```

R Code (Cont'd)

Width	N of Cases	N of Satellites	Sample Mean	Sample Variance
$(-\infty, 23.25]$	14	14	1.00	2.77
$(23.25, 24.25]$	14	20	1.43	8.88
$(24.25, 25.25]$	28	67	2.39	6.54
$(25.25, 26.25]$	39	105	2.69	11.38
$(26.25, 27.25]$	22	63	2.86	6.88
$(27.25, 28.25]$	24	93	3.87	8.81
$(28.25, 29.25]$	18	71	3.94	16.88
$(29.25, \infty]$	14	72	5.14	8.29

```
## Horseshoe crab
crab <- data.frame(
  width_bin = factor(c("(-
Inf,23.25]", "(23.25,24.25]", "(24.25,25.25]", "(25.25,26.25]",
                      "(26.25,27.25]", "(27.25,28.25]", "(28.25,29.25]", "(29.25,Inf]"),
  levels = c("(-
Inf,23.25]", "(23.25,24.25]", "(24.25,25.25]", "(25.25,26.25]",
             "(26.25,27.25]", "(27.25,28.25]", "(28.25,29.25]", "(29.25,Inf]")),
  width_mid = c(22.75, 23.75, 24.75, 25.75, 26.75, 27.75, 28.75, 29.75), # 구간 중앙값(근사)
  Ncases    = c(14, 14, 28, 39, 22, 24, 18, 14),
  Nsat      = c(14, 20, 67, 105, 63, 93, 71, 72)
)

fit_pois <- glm(Nsat ~ width_mid, family = poisson, data = crab)
summary(fit_pois)

# Pearson phi_hat: sum(Pearson^2)/df
pearson_res <- residuals(fit_pois, type="pearson")
phi_hat    <- sum(pearson_res^2) / fit_pois$df.residual
phi_hat    # >1: overdispersion
```

R Code (Cont'd)

Width	N of Cases	N of Satellites	Sample Mean	Sample Variance
$(-\infty, 23.25]$	14	14	1.00	2.77
$(23.25, 24.25]$	14	20	1.43	8.88
$(24.25, 25.25]$	28	67	2.39	6.54
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$(26.25, 27.25]$	22	63	2.86	6.88
$(27.25, 28.25]$	24	93	3.87	8.81
$(28.25, 29.25]$	18	71	3.94	16.88
$(29.25, \infty]$	14	72	5.14	8.29

```
# Offset
fit_pois_off <- glm(Nsat ~ width_mid + offset(log(Ncases)), family = poisson,
data = crab)
summary(fit_pois_off)

pearson_res <- residuals(fit_pois_offset, type="pearson")
phi_hat <- sum(pearson_res^2) / fit_pois$df.residual; phi_hat

# Quasi-Poisson (quasi-likelihood)
fit_qpois <- glm(Nsat ~ width_mid, family=quasipoisson, data=crab) #
Var(Y)=phi*mu
summary(fit_qpois)

# Negative binomial (Poisson-Gamma)
fit_nb <- MASS::glm.nb(Nsat ~ width_mid, data=crab)
summary(fit_nb)

# Poisson vs Negative binomial
lmtest::lrtest(fit_nb, fit_pois)
```

R Code (Cont'd)

<Collisions involving trains in Great Britain>

Year	Train-km	Train Collisions	Train-road Collisions	Year	Train-km	Train Collisions	Train-road Collisions
2003	518	0	3	1988	443	2	4
2002	516	1	3	1987	397	1	6
2001	508	0	4	1986	414	2	13
2000	503	1	3	1985	418	0	5
1999	505	1	2	1984	389	5	3
1998	487	0	4	1983	401	2	7
1997	463	1	1	1982	372	2	3
1996	437	2	2	1981	417	2	2
1995	423	1	2	1980	430	2	2
1994	415	2	4	1979	426	3	3
1993	425	0	4	1978	430	2	4
1992	430	1	4	1977	425	1	8
1991	439	2	6	1976	426	2	12
1990	431	1	2	1975	436	5	2
1989	436	4	4				

Source: British Department of Transport.

```
## British Train Accidents
```

```
train <- data.frame(  
  year = 1975:2003,  
  train_km =  
c(436,426,426,425,430,426,420,372,401,389,418,414,397,443,436,431,43  
9,430,425,415,423,437,463,487,505,503,508,516,518),  
  train_collisions = c(5,2,2,1,2,3,2,2,2,5,0,2,1,2,0,1,2,1,3,2,2,2,1,2,1,0,0,1,0),  
  train_load_collisions =  
c(2,12,8,4,3,2,2,3,7,3,5,13,6,4,4,2,6,4,4,4,2,2,1,4,2,3,4,3,3)  
)
```

```
train$x <- train$year - 1975
```

```
m_off <- glm(train_load_collisions ~ x + offset(log(train_km)), family =  
poisson, data = train)  
summary(m_off)
```